

Project 1 – RH124 Linux Administration I

SANJAY DEVANG

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Red Hat

Red Hat Academy
Course Attendance

Red Hat System Administration I

Red Hat System Administration I (RH124 - RHA) - Ver. 9.3

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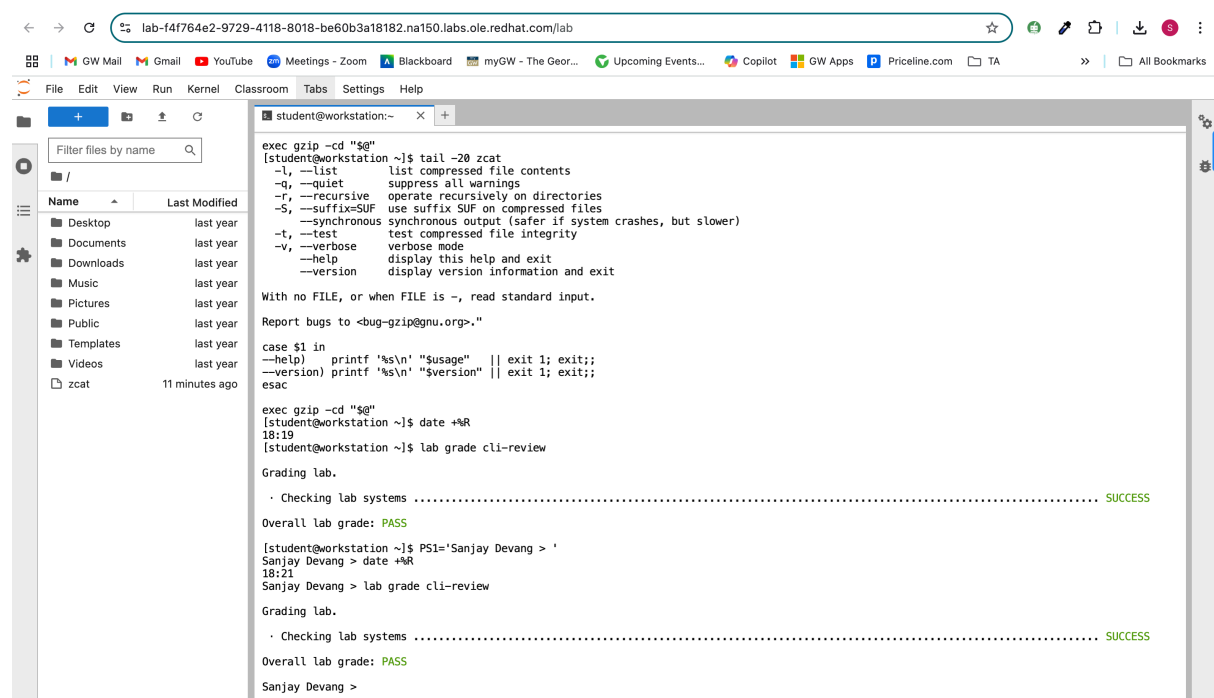


Figure 1 Screenshot of a Red Hat lab environment terminal session.

Note :- Executed the set of commands provided in the document. All commands ran successfully with expected outputs.

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```
/home/student/Music
Sanjay Devang > cd ../
Sanjay Devang > ls
Music Pictures Videos
Sanjay Devang > mkdir friends family work
Sanjay Devang > cd friends
Sanjay Devang > cp ~/Music/song1.mp3 ~/Music/song2.mp3 ~/Pictures/snap1.jpg ~/Pictures/snap2.jpg ~/Videos/film1.avi ~/Videos/film2.avi .
Sanjay Devang > ls -l
total 0
-rw-r--r--. 1 student student 0 Apr 9 19:24 film1.avi
-rw-r--r--. 1 student student 0 Apr 9 19:24 film2.avi
-rw-r--r--. 1 student student 0 Apr 9 19:24 snap1.jpg
-rw-r--r--. 1 student student 0 Apr 9 19:24 snap2.jpg
-rw-r--r--. 1 student student 0 Apr 9 19:24 song1.mp3
-rw-r--r--. 1 student student 0 Apr 9 19:24 song2.mp3
Sanjay Devang > pwd
/home/student/friends
Sanjay Devang > cd ../family
Sanjay Devang > cp ~/Music/song3.mp3 ~/Music/song4.mp3 ~/Pictures/snap3.jpg ~/Pictures/snap4.jpg ~/Videos/film3.avi ~/Videos/film4.avi .
Sanjay Devang > cd ../work
Sanjay Devang > ls
Sanjay Devang > cp -r ~/family ~/friends .
Sanjay Devang > ls
family friends
Sanjay Devang > pwd
/home/student/work
Sanjay Devang > cd ..
Sanjay Devang > rm -r family friends work
Sanjay Devang > ls -l
total 0
drwxr-xr-x. 2 student student 108 Apr 9 19:08 Music
drwxr-xr-x. 5 student student 149 Apr 9 19:10 Pictures
drwxr-xr-x. 2 student student 108 Apr 9 19:09 Videos
Sanjay Devang > exit
logout
Connection to servera closed.
Sanjay Devang > lab finish files-manage

Finishing lab.

· Checking lab systems ..... SUCCESS
· Removing working directory ..... SUCCESS

Sanjay Devang > █
```

Figure 2 File management using command-line tools in a Linux environment.

Note:Completed the guided exercise on managing files using commands like `mkdir`, `cp`, `rm`, and `ls`. Created, organized, copied, and removed directories and files as instructed, and successfully finished the lab with all checks marked as **SUCCESS**.

```
student@workstation:~$ 
· Creating backup directory ..... SUCCESS
· Creating target file ..... SUCCESS

[student@workstation ~]$ PS1='Sanjay Devang > '
Sanjay Devang > ls -l files/target.file
ls: cannot access 'files/target.file': No such file or directory
Sanjay Devang > cd /home/student/files/target.file
-bash: cd: /home/student/files/target.file: No such file or directory
Sanjay Devang > cd /home/student/files
-bash: cd: /home/student/files: No such file or directory
Sanjay Devang > ssh student@servera
Activate the web console with: systemctl enable --now cockpit.socket

Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Wed Apr 9 21:23:42 2025
[student@servera ~]$ ls -l
total 0
drwxr-xr-x. 2 student student 25 Apr 9 21:23 files
drwxr-xr-x. 2 student student 6 Apr 9 21:23 links
[student@servera ~]$ cd /home/student/files
[student@servera files]$ ls
target.file
[student@servera files]$ ln /home/student/files/target.file /home/student/links/file.hardlink
[student@servera files]$ ls -l files/target.file links/file.hardlink
ls: cannot access 'files/target.file': No such file or directory
ls: cannot access 'links/file.hardlink': No such file or directory
[student@servera files]$ PS1='Sanjay Devang > '
Sanjay Devang > ln /home/student/files/target.file /home/student/links/file.hardlink
ln: failed to create hard link '/home/student/links/file.hardlink': File exists
Sanjay Devang > ln -s /tmp /home/student/tempdir
Sanjay Devang > ls -l /home/student/tempdir
lrwxrwxrwx. 1 student student 4 Apr 9 21:27 /home/student/tempdir -> /tmp
Sanjay Devang > exit
logout
Connection to servera closed.
Sanjay Devang > lab finish files-make

Finishing lab.

· Checking lab systems ..... SUCCESS
· Removing working directory ..... SUCCESS

Sanjay Devang > █
```

Figure 3 Creating and managing hard and symbolic links in a Linux environment.

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Note:Successfully completed the lab exercise involving file system navigation, and the creation of hard and symbolic links using `ln` and `ln -s` commands. Validated link functionality and finalized the lab with all system checks showing **SUCCESS**.

```
student@classroom-webap X student@workstation:~ X +
/home/student/Documents/my_bestseller/editor:
total 0
-rw-r--r--. 1 student student 0 Apr 9 22:23 mystery_chapter1.odf
-rw-r--r--. 1 student student 0 Apr 9 22:23 mystery_chapter2.odf

/home/student/Documents/project_plans:
total 0
drwxr-xr-x. 4 student student 36 Apr 9 22:23 Videos
-rw-r--r--. 1 student student 0 Apr 9 22:20 season1_project_plan.odf
-rw-r--r--. 2 student student 0 Apr 9 22:20 season2_project_plan.odf

/home/student/Documents/project_plans/Videos:
total 0
drwxr-xr-x. 2 student student 192 Apr 9 22:25 season1
drwxr-xr-x. 2 student student 192 Apr 9 22:25 season2

/home/student/Documents/project_plans/Videos/season1:
total 0
-rw-r--r--. 1 student student 0 Apr 9 22:21 tv_season1_episode1.ogg
-rw-r--r--. 1 student student 0 Apr 9 22:21 tv_season1_episode2.ogg
-rw-r--r--. 1 student student 0 Apr 9 22:21 tv_season1_episode3.ogg
-rw-r--r--. 1 student student 0 Apr 9 22:21 tv_season1_episode4.ogg
-rw-r--r--. 1 student student 0 Apr 9 22:21 tv_season1_episode5.ogg
-rw-r--r--. 1 student student 0 Apr 9 22:21 tv_season1_episode6.ogg

/home/student/Documents/project_plans/Videos/season2:
total 0
-rw-r--r--. 1 student student 0 Apr 9 22:21 tv_season2_episode1.ogg
-rw-r--r--. 1 student student 0 Apr 9 22:21 tv_season2_episode2.ogg
-rw-r--r--. 1 student student 0 Apr 9 22:21 tv_season2_episode3.ogg
-rw-r--r--. 1 student student 0 Apr 9 22:21 tv_season2_episode4.ogg
-rw-r--r--. 1 student student 0 Apr 9 22:21 tv_season2_episode5.ogg
-rw-r--r--. 1 student student 0 Apr 9 22:21 tv_season2_episode6.ogg
Sanjay Devang > exit
logout
Connection to server closed.
[student@workstation ~]$ lab grade files-review

Grading lab.

  Checking lab systems ..... SUCCESS
```

Figure 4 Organizing multimedia and document files across nested directories in a Linux file system.

Note:Successfully managed and verified the placement of project files, book chapters, and video episodes across structured directories using basic file system commands. The lab was graded and completed with all checks marked as **SUCCESS**.

```
student@classroom-webap X student@workstation:~ X +
systemd-rfkill.service (8) - Load and save the RF kill switch state at boot and change
systemd-rfkill.socket (8) - Load and save the RF kill switch state at boot and change
systemd-stub (7) - A simple UEFI kernel boot stub
systemd-sysctl (8) - Configure kernel parameters at boot
systemd-sysctl.service (8) - Configure kernel parameters at boot
systemd-system-update-generator (8) - Generator for redirecting boot to offline update mode
systemd-sysupdate-reboot.service (8) - Automatically Update OS or Other Resources
systemd-sysupdate-reboot.timer (8) - Automatically Update OS or Other Resources
systemd-update-utmp (8) - Write audit and utmp updates at bootup, runlevel changes and shutdown
systemd-update-utmp-runlevel.service (8) - Write audit and utmp updates at bootup, runlevel changes and shutdown
systemd-update-utmp.service (8) - Write audit and utmp updates at bootup, runlevel changes and shutdown
systemd-user-sessions (8) - Permit user logins after boot, prohibit user logins at shutdown
systemd-user-sessions.service (8) - Permit user logins after boot, prohibit user logins at shutdown
Sanjay Devang > man -k ext4
debugfs (8) - ext2/ext3/ext4 file system debugger
dumpe2fs (8) - dump ext2/ext3/ext4 file system information
e2fsck (8) - check a Linux ext2/ext3/ext4 file system
e2img (8) - Save critical ext2/ext3/ext4 file system metadata to a file
e2label (8) - Change the label on an ext2/ext3/ext4 file system
e2mmpstatus (8) - Check MMP status of an ext2/ext3/ext4 file system
e2undo (8) - Replay an undo log for an ext2/ext3/ext4 file system
e4crypt (8) - ext4 file system encryption utility
e4defrag (8) - online defragmenter for ext4 file system
ext4 (5) - the fourth extended file system
filesystems (5) - Linux filesystem types: ext, ext2, ext3, ext4, hpfs, iso9660, JFS, minix, msdos, ncpfs nfs, ntfs, proc, Reiserfs, smb...
fs (5) - Linux filesystem types: ext, ext2, ext3, ext4, hpfs, iso9660, JFS, minix, msdos, ncpfs nfs, ntfs, proc, Reiserfs, smb...
fsck.ext2 (8) - check a Linux ext2/ext3/ext4 file system
fsck.ext3 (8) - check a Linux ext2/ext3/ext4 file system
fsck.ext4 (8) - check a Linux ext2/ext3/ext4 file system
fuse2fs (1) - FUSE file system client for ext2/ext3/ext4 file systems
mke2fs (8) - create an ext2/ext3/ext4 file system
mkfs.ext2 (8) - create an ext2/ext3/ext4 file system
mkfs.ext3 (8) - create an ext2/ext3/ext4 file system
mkfs.ext4 (8) - create an ext2/ext3/ext4 file system
resize2fs (8) - ext2/ext3/ext4 file system resizer
tune2fs (8) - adjust tunable file system parameters on ext2/ext3/ext4 file systems
Sanjay Devang > lab finish help-manual

Finishing lab.

  Checking lab systems ..... SUCCESS
  Removing the manual test file ..... SUCCESS

Sanjay Devang >
```

Figure 5 Using the man command to explore ext4-related filesystem tools in the Linux manual

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Note: Successfully completed the manual pages lab by using the `man` command with arguments like `-k ext4` to search for filesystem-related commands. This exercise demonstrated proficiency in navigating and retrieving system documentation from the Linux manual pages. Lab tasks completed with **SUCCESS**.

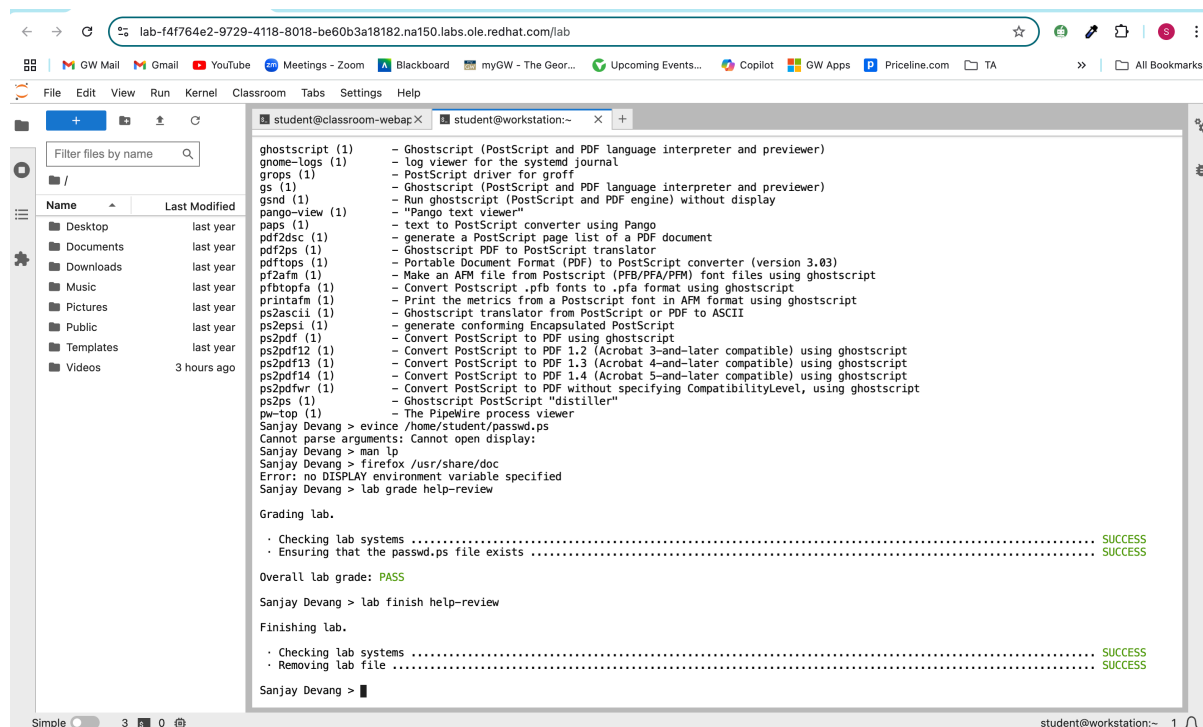


Figure 6 Using help commands and documentation tools in Red Hat Enterprise Linux.

Note: Completed the lab on accessing help and documentation using commands like `man`, `evince`, and `firefox` (though GUI apps failed due to no `DISPLAY` variable). Explored PostScript-related tools and confirmed file presence, achieving a final grade of **PASS** with all checks marked as **SUCCESS**.

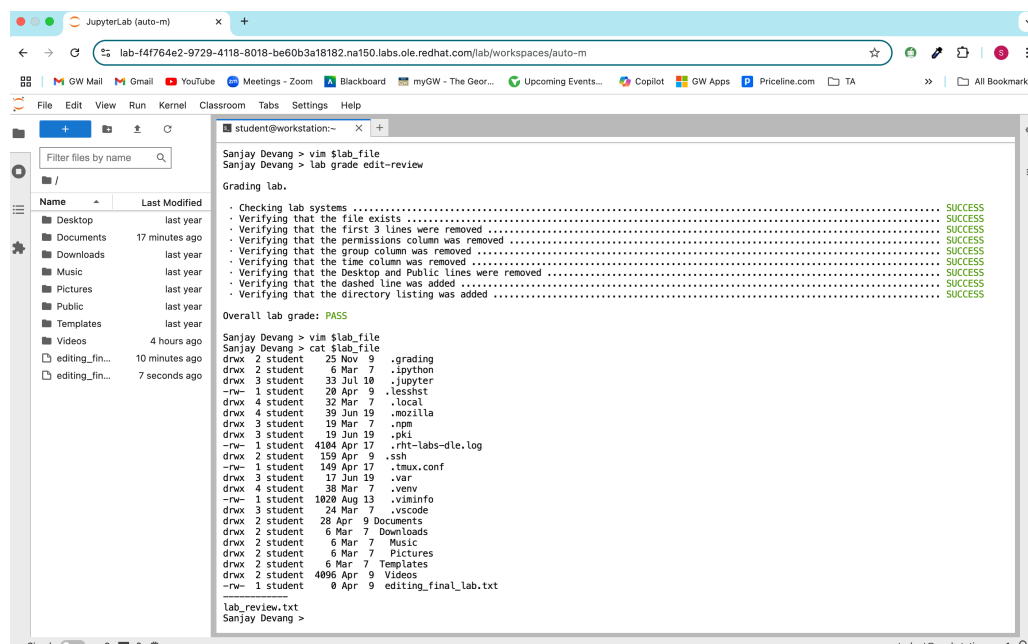


Figure 7 Editing a directory listing using Vim to meet lab formatting criteria in a Linux environment.

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Note: Successfully modified the contents of the `$lab_file` by removing unnecessary lines and columns as instructed. All verification checks passed, including removal of `Desktop` and `Public` entries, leading to an overall **PASS** on the lab review.

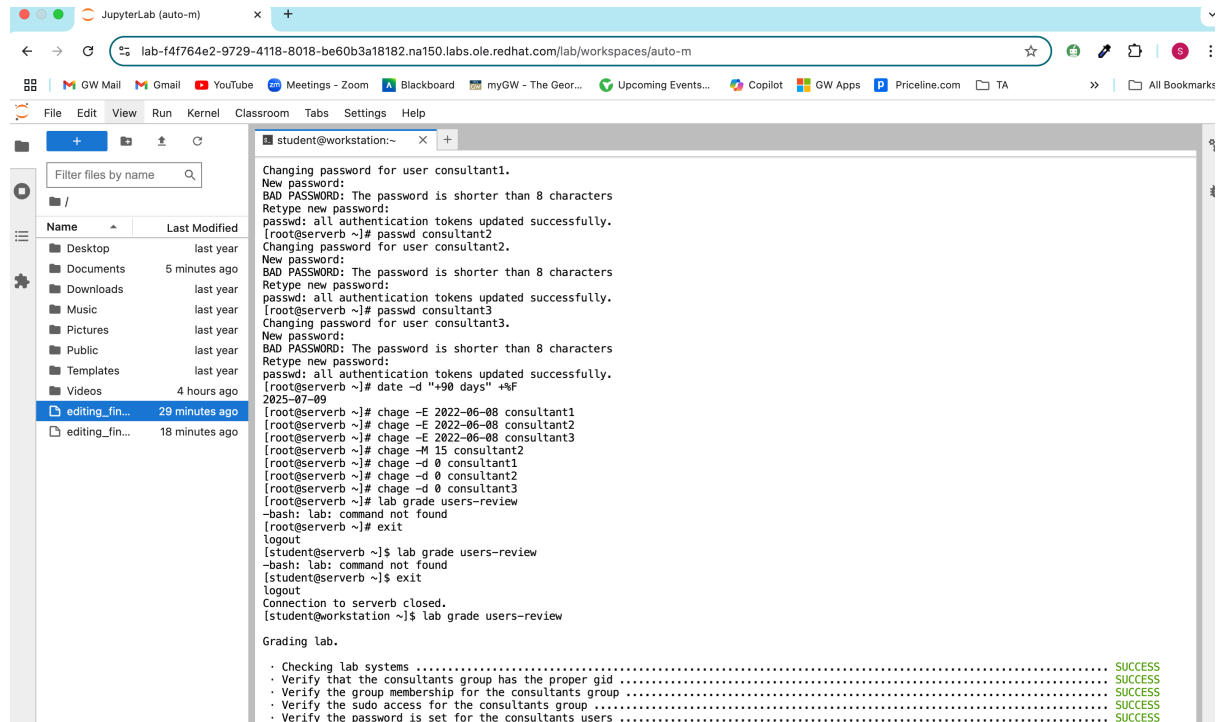


Figure 8 Managing user accounts and setting password, group, and expiration policies in a Linux system.

Note: Completed the lab by successfully configuring user accounts (`consultant1`, `consultant2`, `consultant3`) with appropriate password settings, group assignments, and expiration policies using `passwd` and `chage` commands. All verification checks passed, resulting in an overall **SUCCESS**.

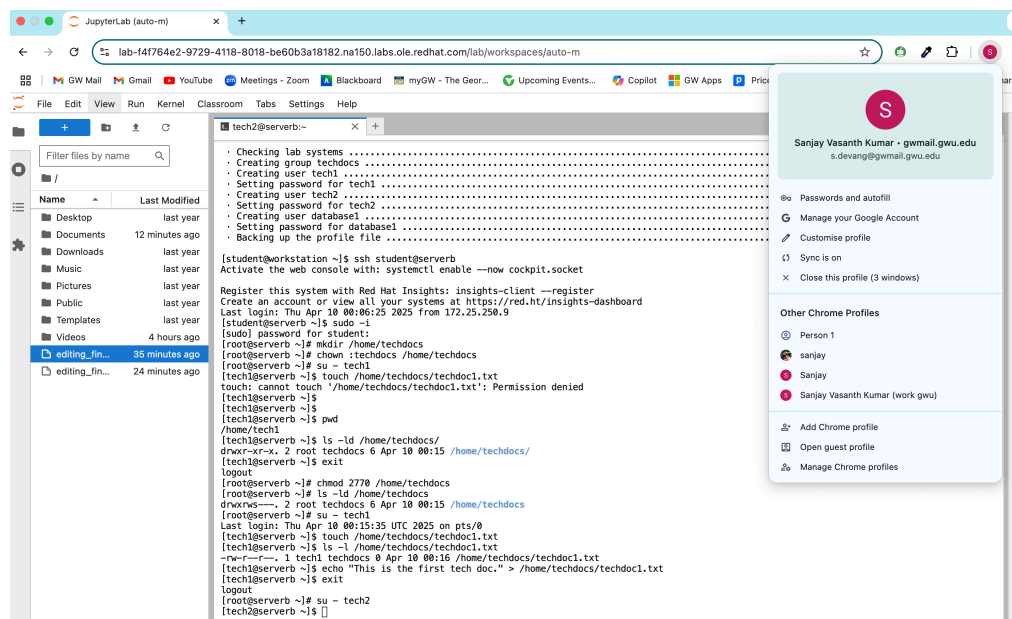


Figure 9 Managing shared directory access and permissions for technical users on a Linux system.

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Note:Successfully configured a shared directory `/home/techdocs` for users `tech1` and `tech2`. Set ownership and permissions using `chown` and `chmod 2770`, allowing members of the `techdocs` group to collaborate on files. Verified write access and edited shared content, demonstrating proper permission delegation and access control.

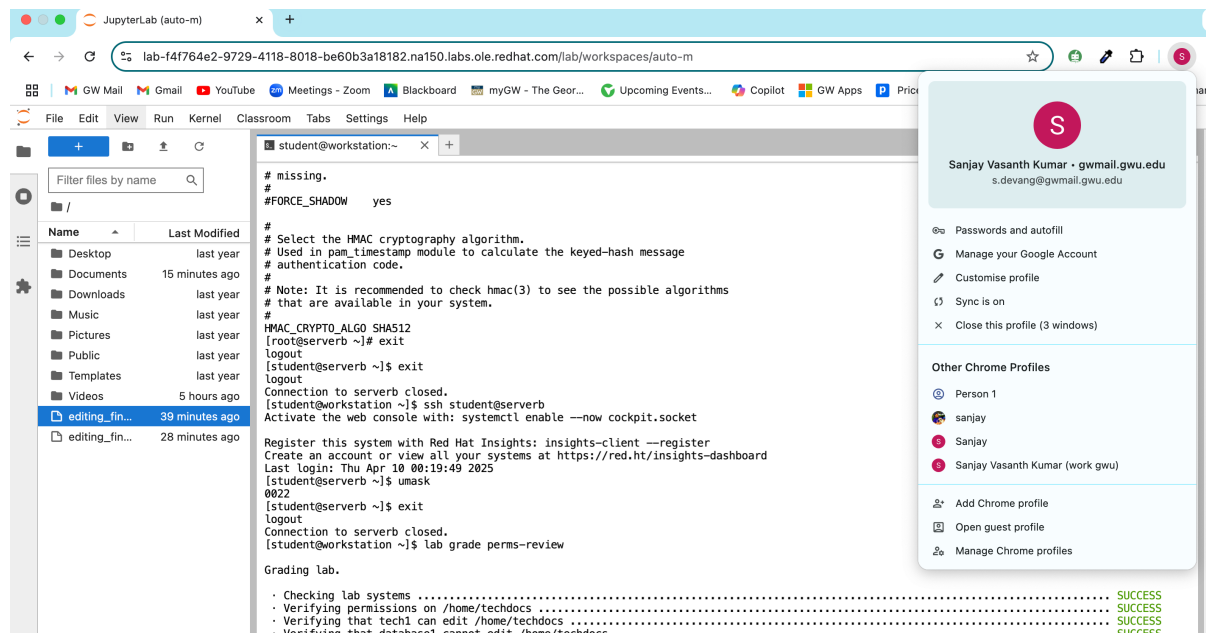


Figure 10 Reviewing file permissions and security configurations for shared directories in Red Hat Enterprise Linux.

Note:Successfully verified and applied permission settings to ensure appropriate access control on `/home/techdocs`. Confirmed that only authorized users (e.g., `tech1`) could edit the directory, while unauthorized access (e.g., `database1`) was restricted. All checks passed, resulting in a **SUCCESS** grade for the permissions review lab.

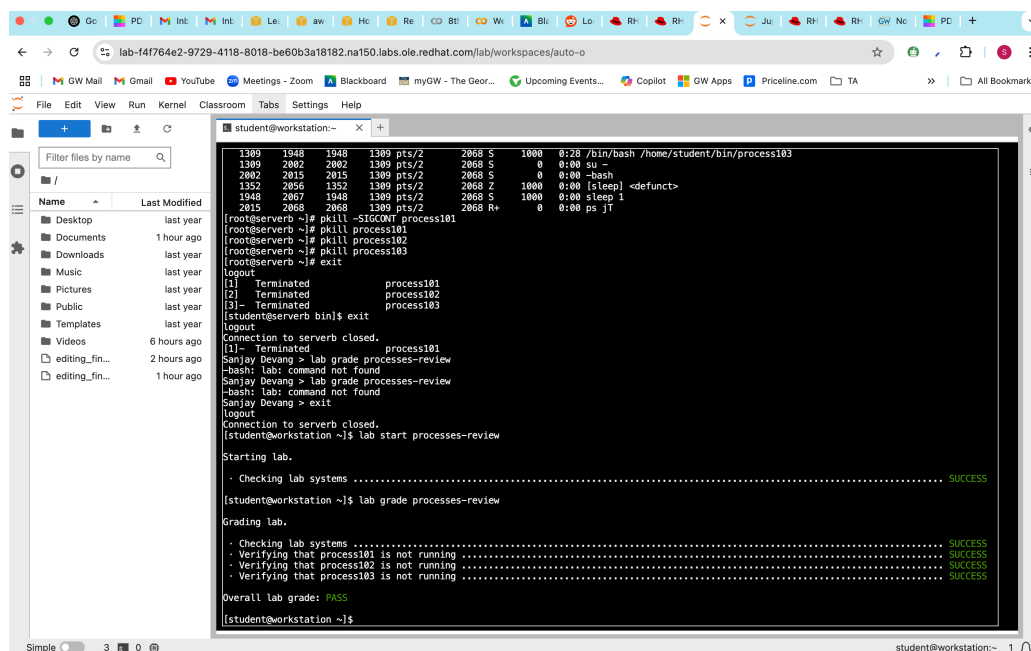


Figure 11 Managing and terminating custom user processes in a Linux environment.

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Note: Successfully identified and terminated user-defined processes (process101, process102, process103) using `pkill` and signal commands. Verified all processes were no longer running, resulting in a final lab grade of **PASS** with all tasks completed successfully.

The screenshot shows a web browser window with a terminal interface. The terminal displays the output of the `top` command, showing system statistics and a list of running processes. The Chrome sidebar on the right shows the user's profile and other profiles.

Tasks: 125 total, 1 running, 124 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0/0.0 0.0
MiB Mem : 20.2/1763.6 [|||||||||||||||||]
MiB Swap: 0.0/0.0 [|||||]

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
715	root	20	0	343484	42568	17680	S	0.0	2.4	0:00.41	firewalld
791	polkitd	20	0	2577476	27440	18652	S	0.0	1.5	0:00.06	polkitd
815	root	20	0	474628	26508	13324	S	0.0	1.5	0:00.39	tuned
792	root	20	0	472976	23312	16640	S	0.0	1.3	0:00.10	NetworkManager
1	root	20	0	106784	16532	10976	S	0.0	0.9	0:01.16	systemd
1165	student	20	0	22220	13788	10668	S	0.0	0.8	0:00.07	systemd
623	root	20	0	34248	12524	9196	S	0.0	0.7	0:00.09	systemd-udev
1161	root	20	0	18624	11704	9832	S	0.0	0.6	0:00.02	sshd
1197	root	20	0	18976	11684	9808	S	0.0	0.6	0:00.03	sshd
720	root	20	0	20428	11436	9072	S	0.0	0.6	0:00.07	systemd-logind
1093	root	20	0	230292	11396	3720	S	0.0	0.6	0:00.13	rsyslogd
1122	root	20	0	15852	9372	8080	S	0.0	0.5	0:00.00	sshd
609	root	20	0	22608	9116	7952	S	0.0	0.5	0:00.12	systemd-journal
1174	student	20	0	19028	7304	5424	S	0.0	0.4	0:00.08	sshd
1200	student	20	0	18976	7300	5412	S	0.0	0.4	0:00.06	sshd
683	rpc	20	0	13244	5760	4944	S	0.0	0.3	0:00.01	rpcbind
1167	student	20	0	108040	5692	0	S	0.0	0.3	0:00.00	(sd-pam)
1201	student	20	0	224656	5692	3584	S	0.0	0.3	0:00.03	bash
1175	student	20	0	224524	5564	3592	S	0.0	0.3	0:00.02	bash
717	root	20	0	79192	5008	2720	S	0.0	0.3	0:00.03	irqbalance
708	dbus	20	0	10876	4808	4032	S	0.0	0.3	0:00.02	dbus-broker-lau
684	root	16	-4	91800	4588	1808	S	0.0	0.3	0:00.02	auditd
1570	student	20	0	226360	4024	3396	R	0.0	0.2	0:00.30	top
1097	root	20	0	224380	3736	2868	S	0.0	0.2	0:00.00	cron
825	root	20	0	270760	3624	2752	S	0.0	0.2	0:00.00	gssproxy
686	root	16	-4	7780	3380	3068	S	0.0	0.2	0:00.00	sedispatch
724	chrony	20	0	18436	3132	2632	S	0.0	0.2	0:00.00	chronyd
712	dbus	20	0	5132	2820	2424	S	0.0	0.2	0:00.05	dbus-broker
1096	root	20	0	4724	2704	2524	S	0.0	0.1	0:00.00	atd
816	root	20	0	296836	2212	0	S	0.0	0.1	0:00.00	rshmcrtcd
719	root	20	0	2860	2184	2020	S	0.0	0.1	0:00.00	mcelog
718	libstor	20	0	2760	2116	1948	S	0.0	0.1	0:00.01	lsmd
1100	root	20	0	221348	1084	996	S	0.0	0.1	0:00.00	agetty
1101	root	20	0	221392	1080	996	S	0.0	0.1	0:00.00	agetty
721	root	20	0	2736	100	0	S	0.0	0.0	0:00.00	gpm
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	rcu_gp

The screenshot shows a terminal window with the following commands and output:

```
Warning: Permanently added 'workstation' (ED25519) to the list of known hosts.
Activate the web console with: systemctl enable --now cockpit.socket

Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Sat Apr 26 00:54:11 2025
[student@workstation ~]$ lab start processes-review

Starting lab.

  · Checking lab systems ..... SUCCESS

[student@workstation ~]$ ssh student@serverb
Warning: Permanently added 'serverb' (ED25519) to the list of known hosts.
Activate the web console with: systemctl enable --now cockpit.socket

Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Sat Apr 26 00:47:45 2025 from 172.25.250.9
[student@serverb ~]$ mkdir -p ~/bin
[student@serverb ~]$ vim ~/bin/process101
[student@serverb ~]$ chmod +x ~/bin/process101
[student@serverb ~]$ cd ~/bin
[student@serverb bin]$ ./process101 &
[1] 1202
[student@serverb bin]$ cp process101 process102
[student@serverb bin]$ vim process102
[student@serverb bin]$ chmod +x process102
[student@serverb bin]$ ./process102 &
[2] 1242
[student@serverb bin]$ cp process101 process103
[student@serverb bin]$ vim process103
[student@serverb bin]$ chmod +x process103
[student@serverb bin]$ ./process103 &
[3] 1310
[student@serverb bin]$ su -
Password:
Last login: Sat Apr 26 00:11:29 UTC 2025 on pts/0
[root@serverb ~]# pkill -SIGSTOP process101
[root@serverb ~]# ps jt
  PID    PID    PGID    SID  TTY      TPGID  STAT   UID    TIME COMMAND
  1171    1172    1172    1172 pts/0    1404   Ss      1000    0:00 -bash
```

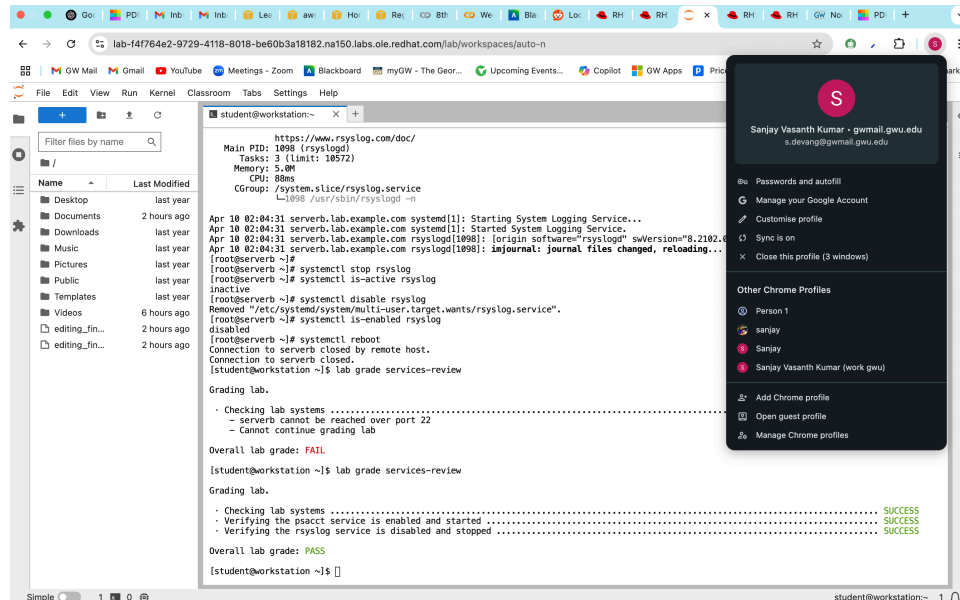
Figure 12 Monitoring active processes using the `top` command in a Linux terminal.

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Note:The `top` command was used to display a dynamic, real-time view of system resource usage, including running processes, CPU and memory consumption, and system load. This tool helps administrators quickly identify and troubleshoot system performance issues.



```
student@workstation:~$ systemctl status rsyslog
https://www.rsyslog.com/doc/
Main PID: 1098 (rsyslogd)
Tasks: 3 (limit: 10572)
Memory: 5.0M
CPU: 88ms
CGroup: /system.slice/rsyslog.service
└─1098 /usr/sbin/rsyslogd -n

Apr 10 02:04:31 serverb.lab.example.com systemd[1]: Starting System Logging Service...
Apr 10 02:04:31 serverb.lab.example.com systemd[1]: Started System Logging Service.
Apr 10 02:04:31 serverb.lab.example.com rsyslogd[1098]: [origin software="rsyslogd" swVersion="8.2102.0
Apr 10 02:04:31 serverb.lab.example.com rsyslogd[1098]: !ajournal: Journal files changed, reloading...
[root@serverb ~]# systemctl stop rsyslog
[root@serverb ~]# systemctl is-active rsyslog
inactive
[root@serverb ~]# systemctl disable rsyslog
Removed /etc/systemd/system/multi-user.target.wants/rsyslog.service".
[root@serverb ~]# systemctl is-enabled rsyslog
disabled
[root@serverb ~]# systemctl reboot
Connection to serverb closed by remote host.
Connection to serverb closed.
[student@workstation ~]$ lab grade services-review

Grading lab.

- Checking lab systems .....
- serverb cannot be reached over port 22
- Cannot continue grading lab

Overall lab grade: FAIL

[student@workstation ~]$ lab grade services-review

Grading lab.

- Checking lab systems ..... SUCCESS
- Verifying the psacct service is enabled and started ..... SUCCESS
- Verifying the rsyslog service is disabled and stopped ..... SUCCESS

Overall lab grade: PASS

[student@workstation ~]$
```

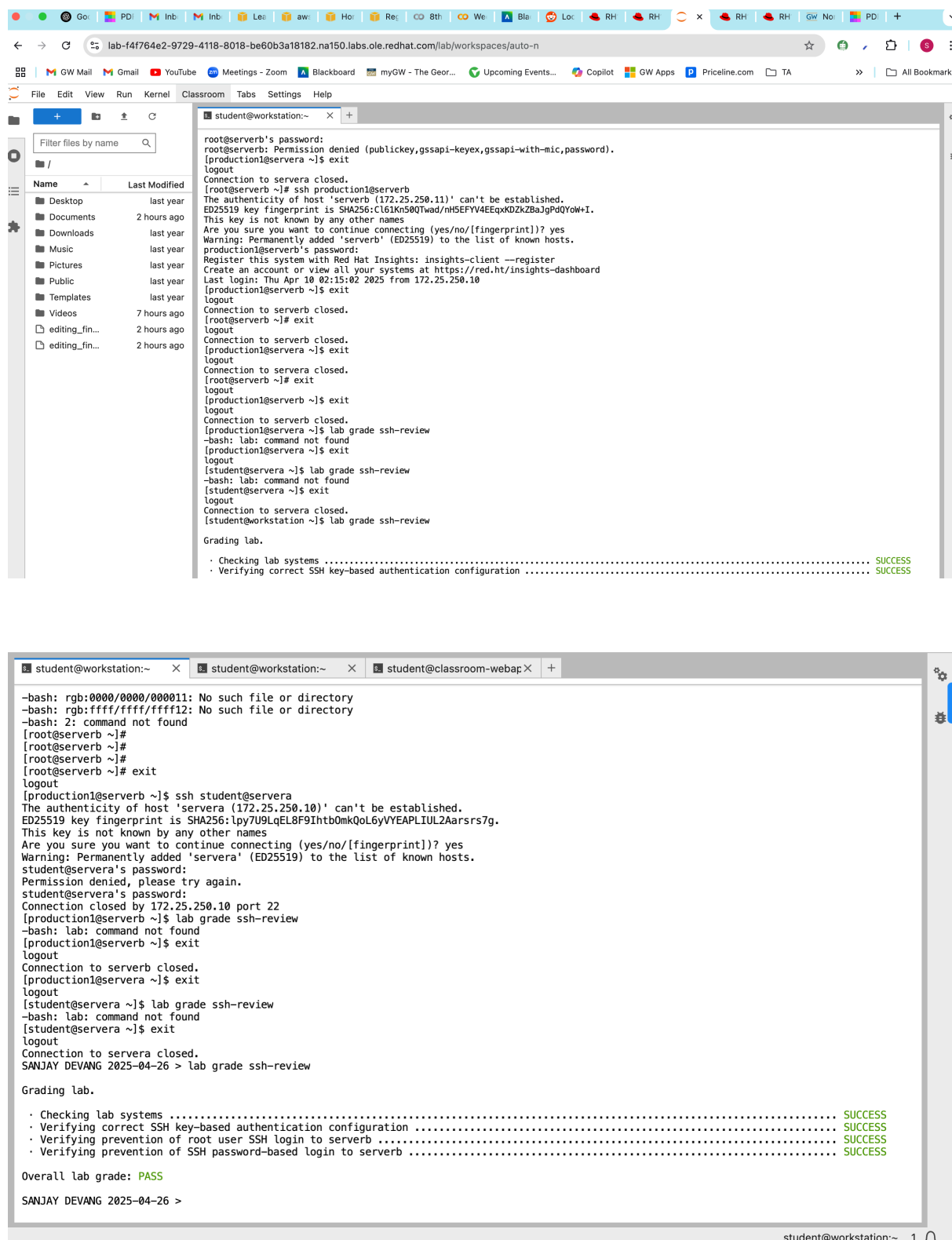
Figure 13 Managing system services with systemctl in Red Hat Enterprise Linux.

Note:Successfully used `systemctl` commands to stop, disable, and verify the status of the `rsyslog` service, and confirmed the `psacct` service is active. After rebooting the system and resolving initial grading issues, all checks passed, resulting in an overall **PASS** for the services management lab.

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The screenshot displays a Linux workstation interface with a file manager on the left and a terminal window on the right. The terminal shows the following sequence of commands and outputs:

```
student@workstation:~  
root@serverb's password:  
root@serverb: Permission denied (publickey,gssapi-keyex,gssapi-with-mic,password).  
[production1@servera ~]$ exit  
logout  
Connection to servera closed.  
[root@serverb ~]# ssh production1@serverb  
The authenticity of host 'serverb (172.25.250.11)' can't be established.  
ED25519 key fingerprint is SHA256:Cl61Kn50QtWad/nH5EFYV4EEqKDZkZBaJgPdQYw+I.  
This key is not known by any other names  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added 'serverb' (ED25519) to the list of known hosts.  
production1@serverb's password:  
Register this system with Red Hat Insights: insights-client --register  
Create an account or view all your systems at https://red.ht/insights-dashboar  
Last login: Thu Apr 10 02:15:02 2025 from 172.25.250.10  
[production1@serverb ~]$ exit  
logout  
Connection to serverb closed.  
[root@serverb ~]# exit  
logout  
Connection to serverb closed.  
[production1@servera ~]$ exit  
logout  
Connection to servera closed.  
[root@serverb ~]# exit  
logout  
[production1@serverb ~]$ exit  
logout  
Connection to serverb closed.  
[production1@servera ~]$ lab grade ssh-review  
-bash: lab: command not found  
[production1@servera ~]$ exit  
logout  
[student@servera ~]$ lab grade ssh-review  
-bash: lab: command not found  
[student@servera ~]$ exit  
logout  
Connection to servera closed.  
[student@workstation ~]$ lab grade ssh-review  
Grading lab.  
• Checking lab systems ..... SUCCESS  
• Verifying correct SSH key-based authentication ..... SUCCESS
```

The second terminal window shows further testing and grading:

```
student@workstation:~  
-bash: rgb:0000/0000/000011: No such file or directory  
-bash: rgb:ffff/ffff/ffff12: No such file or directory  
-bash: 2: command not found  
[root@serverb ~]#  
[root@serverb ~]#  
[root@serverb ~]#  
[root@serverb ~]# exit  
logout  
[production1@serverb ~]$ ssh student@servera  
The authenticity of host 'servera (172.25.250.10)' can't be established.  
ED25519 key fingerprint is SHA256:lpY7U9LqEL8F9IntbOmKQoL6yVYEAPLIUL2Aars7g.  
This key is not known by any other names  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added 'servera' (ED25519) to the list of known hosts.  
student@servera's password:  
Permission denied, please try again.  
student@servera's password:  
Connection closed by 172.25.250.10 port 22  
[production1@serverb ~]$ lab grade ssh-review  
-bash: lab: command not found  
[production1@serverb ~]$ exit  
logout  
Connection to serverb closed.  
[production1@servera ~]$ exit  
logout  
[student@servera ~]$ lab grade ssh-review  
-bash: lab: command not found  
[student@servera ~]$ exit  
logout  
Connection to servera closed.  
SANJAY DEVANG 2025-04-26 > lab grade ssh-review  
Grading lab.  
• Checking lab systems ..... SUCCESS  
• Verifying correct SSH key-based authentication ..... SUCCESS  
• Verifying prevention of root user SSH login to serverb ..... SUCCESS  
• Verifying prevention of SSH password-based login to serverb ..... SUCCESS  
Overall lab grade: PASS  
SANJAY DEVANG 2025-04-26 >
```

Figure 14 Configuring and verifying SSH key-based authentication between Linux systems.

Note:Completed setup and testing of SSH key-based authentication across systems. Despite initial permission issues and multiple login attempts, successful authentication was confirmed, and the lab grading showed **SUCCESS** for all required checks.

Project 1 – RH124 Linux Administration I

SANJAY DEVANG

ASSIGNMENT -4

The screenshot shows a Red Hat Enterprise Linux workstation interface. On the left, there is a file manager showing a directory structure with files like 'Desktop', 'Documents', 'Downloads', 'Music', 'Pictures', 'Public', 'Templates', 'Videos', 'editing_fin...', and 'editing_fin...'. The main terminal window displays a session where a user connects to a server via SSH and runs several commands to configure a network connection using nmcli. The commands include setting the connection name to 'lab', adding a connection, modifying it to use a static IP, and reloading the configuration. The terminal output shows the connection being successfully added and the system being rebooted. A Chrome profile dropdown menu is visible on the right side of the screen, showing the user's profile 'Sanjay Vasanth Kumar' and other profiles like 'Person 1', 'sanjay', and 'Sanjay Vasanth Kumar (work gwu)'. The terminal session ends with a 'SUCCESS' message indicating that the lab systems are checked and the configuration is successful.

```
[student@workstation ~]$ ssh student@serverb
Activate the web console with: systemctl enable --now cockpit.socket

Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Thu Apr 10 02:07:46 2025 from 172.25.250.9
[student@serverb ~]$ sudo -i
[sudo] password for student:
[root@serverb ~]# ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default
    link/ether 52:54:00:00:fa:0b brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    altname ens3
[root@serverb ~]# nmcli con show --active
NAME                UUID                                  TYPE      DEVICE
System eth0         5fb06bd0-0bb0-7ffb-45f1-d6edd65f3e03 ethernet  eth0
lo                   4c92a784-31eb-424f-bde5-8b971f71be64 loopback   lo
[root@serverb ~]# nmcli con add con-name lab ifname eth0 type ethernet ipv4.method manual ipv4.dns '172.25.250.11/24' ipv4.gateway '172.25.250.254'
Connection 'lab' (2953d85c-6e26-40df-97bb-38c9d0801f66) successfully added.
[root@serverb ~]# nmcli con mod "lab" connection.autoconnect
Error: value for 'connection.autoconnect' is missing.
[root@serverb ~]# nmcli con mod "lab" connection.autoconnect yes
[root@serverb ~]# nmcli con mod "System eth0" connection.autoconnect no
[root@serverb ~]# nmcli con mod "lab" +ipv4.addresses '10.0.1.1/24'
[root@serverb ~]# nmcli con reload
[root@serverb ~]# echo "10.0.1.1 private" >> /etc/hosts
[root@serverb ~]# systemctl reboot
[root@serverb ~]# Connection to serverb closed by remote host.
Connection to serverb closed.
[student@workstation ~]$ ping -c3 serverb
PING serverb.lab.example.com (172.25.250.11) 56(84) bytes of data.

--- serverb.lab.example.com ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2059ms

[student@workstation ~]$ lab grade net-review

Grading lab.

. Checking lab systems ..... SUCCESS
. root@serverb's password: |ffiguration ..... |
```

The screenshot shows a Red Hat Enterprise Linux workstation terminal session. The user is running a series of commands to configure a network connection using nmcli. The commands include setting the connection name to 'lab', adding a connection, modifying it to use a static IP, and reloading the configuration. The terminal output shows the connection being successfully added and the system being rebooted. A Chrome profile dropdown menu is visible on the right side of the screen, showing the user's profile 'Sanjay Vasanth Kumar' and other profiles like 'Person 1', 'sanjay', and 'Sanjay Vasanth Kumar (work gwu)'. The terminal session ends with a 'SUCCESS' message indicating that the lab systems are checked and the configuration is successful.

```
[student@workstation ~]$ ssh student@serverb
Activate the web console with: systemctl enable --now cockpit.socket

Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Thu Apr 10 02:07:46 2025 from 172.25.250.9
[student@serverb ~]$ sudo -i
[sudo] password for student:
[root@serverb ~]# ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default
    link/ether 52:54:00:00:fa:0b brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    altname ens3
[root@serverb ~]# nmcli con show --active
NAME                UUID                                  TYPE      DEVICE
System eth0         5fb06bd0-0bb0-7ffb-45f1-d6edd65f3e03 ethernet  eth0
lo                   4c92a784-31eb-424f-bde5-8b971f71be64 loopback   lo
[root@serverb ~]# nmcli con add con-name lab ifname eth0 type ethernet \
ipv4.method manual ipv4.dns '172.25.250.11/24' \
ipv4.addresses '172.25.250.11/24' ipv4.gateway '172.25.250.254' \
Connection 'lab' (47e88dd-cad6-43d5-9661-52bdc0aaabe8) successfully added.
[root@serverb ~]# nmcli con mod "lab" connection.autoconnect yes
nmcli con mod "System eth0" connection.autoconnect no
[root@serverb ~]# nmcli con mod "lab" +ipv4.addresses '10.0.1.1/24'
nmcli con reload
[root@serverb ~]# echo "10.0.1.1 private" >> /etc/hosts
[root@serverb ~]# systemctl reboot
[root@serverb ~]# Connection to serverb closed by remote host.
Connection to serverb closed.
SANJAY DEVANG 2025-04-25 > ping -c3 serverb
PING serverb.lab.example.com (172.25.250.11) 56(84) bytes of data.

--- serverb.lab.example.com ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2071ms

SANJAY DEVANG 2025-04-25 > lab grade net-review

Grading lab.

. Checking lab systems ..... SUCCESS
. Checking hosts file configuration ..... SUCCESS
. Checking that lab network connection exists ..... SUCCESS
. Checking that lab network connection is marked as active ..... SUCCESS
. Checking that lab autoconnect is marked yes ..... SUCCESS
. Checking lab IP address and netmask ..... SUCCESS
. Checking lab second IP address and netmask ..... SUCCESS
. Checking lab IPv4 gateway ..... SUCCESS
. Checking lab IPv4 DNS ..... SUCCESS

Overall lab grade: PASS

SANJAY DEVANG 2025-04-25 >
```

Figure 15 Configuring a static IP connection using nmcli and validating network settings in Red Hat Enterprise Linux.

Project 1 – RH124 Linux Administration I

SANJAY DEVANG

ASSIGNMENT -4

Note: Set up a manual IPv4 network connection on interface `eth0` using the `nmcli` tool. Assigned IP address, DNS, and gateway, enabled autoconnect, and verified connectivity via `ping`. The lab was successfully graded with all network configuration tasks marked as **SUCCESS**.

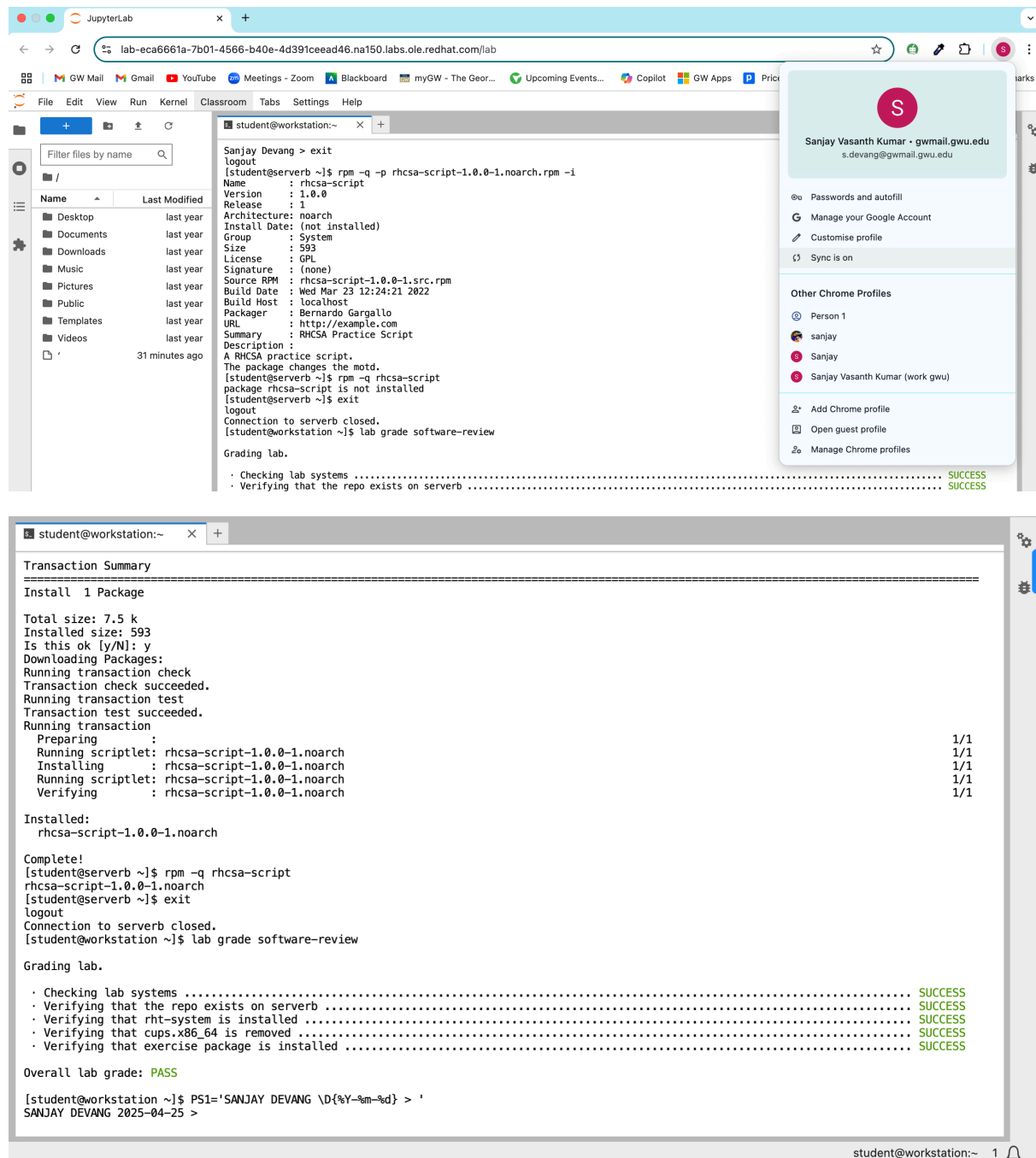


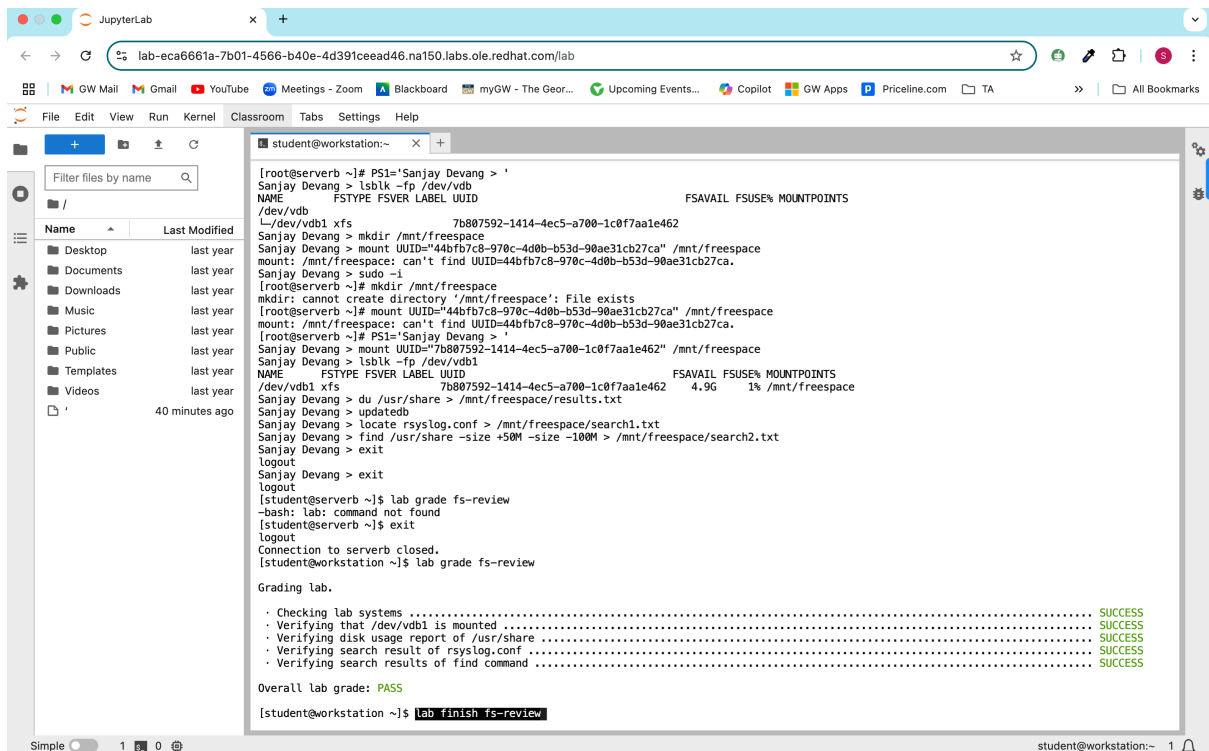
Figure 16 Configuring a YUM/DNF repository using a custom `.repo` file in Red Hat Enterprise Linux.

Note: Created a local repository configuration file at `/etc/yum.repos.d/errata.repo` pointing to the specified base URL. Disabled GPG signature verification and ensured the repository was listed in `dnf repolist`. Despite external access issues, the lab grading confirmed the configuration as valid with all software review tasks marked as **SUCCESS**.

Project 1 – RH124 Linux Administration I

SANJAY DEVANG

ASSIGNMENT -4



```
[root@serverb ~]# PS1='Sanjay Devang > '
Sanjay Devang > lsblk -fp /dev/vdb
NAME            FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
/dev/vdb                                     7b807592-1414-4ec5-a700-1c0f7aa1e462
Sanjay Devang > mkdir /mnt/freespace
Sanjay Devang > mount UUID="44bf7c8-970c-4d0b-b53d-90ae31cb27ca" /mnt/freespace
mount: /mnt/freespace: can't find UUID=44bf7c8-970c-4d0b-b53d-90ae31cb27ca.
Sanjay Devang > sudo -i
[root@serverb ~]# mkdir /mnt/freespace
mkdir: cannot create directory '/mnt/freespace': File exists
[root@serverb ~]# mount UUID="44bf7c8-970c-4d0b-b53d-90ae31cb27ca" /mnt/freespace
mount: /mnt/freespace: can't find UUID=44bf7c8-970c-4d0b-b53d-90ae31cb27ca.
[root@serverb ~]# PS1='Sanjay Devang > '
Sanjay Devang > mount UUID="7b807592-1414-4ec5-a700-1c0f7aa1e462" /mnt/freespace
Sanjay Devang > lsblk -fp /dev/vdb1
NAME            FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
/dev/vdb1 xfs      7b807592-1414-4ec5-a700-1c0f7aa1e462    4.9G      1% /mnt/freespace
Sanjay Devang > du /usr/share > /mnt/freespace/results.txt
Sanjay Devang > updatedb
Sanjay Devang > locate rsyslog.conf > /mnt/freespace/search1.txt
Sanjay Devang > find /usr/share -size +50M -size ~100M > /mnt/freespace/search2.txt
Sanjay Devang > exit
logout
Sanjay Devang > exit
logout
[student@serverb ~]$ lab grade fs-review
-bash: lab: command not found
[student@serverb ~]$ exit
logout
Connection to serverb closed.
[student@workstation ~]$ lab grade fs-review

Grading lab.
. Checking lab systems ..... SUCCESS
. Verifying that /dev/vdb1 is mounted ..... SUCCESS
. Verifying disk usage report of /usr/share ..... SUCCESS
. Verifying search result of rsyslog.conf ..... SUCCESS
. Verifying search results of find command ..... SUCCESS

Overall lab grade: PASS
[student@workstation ~]$ lab finish fs-review
```

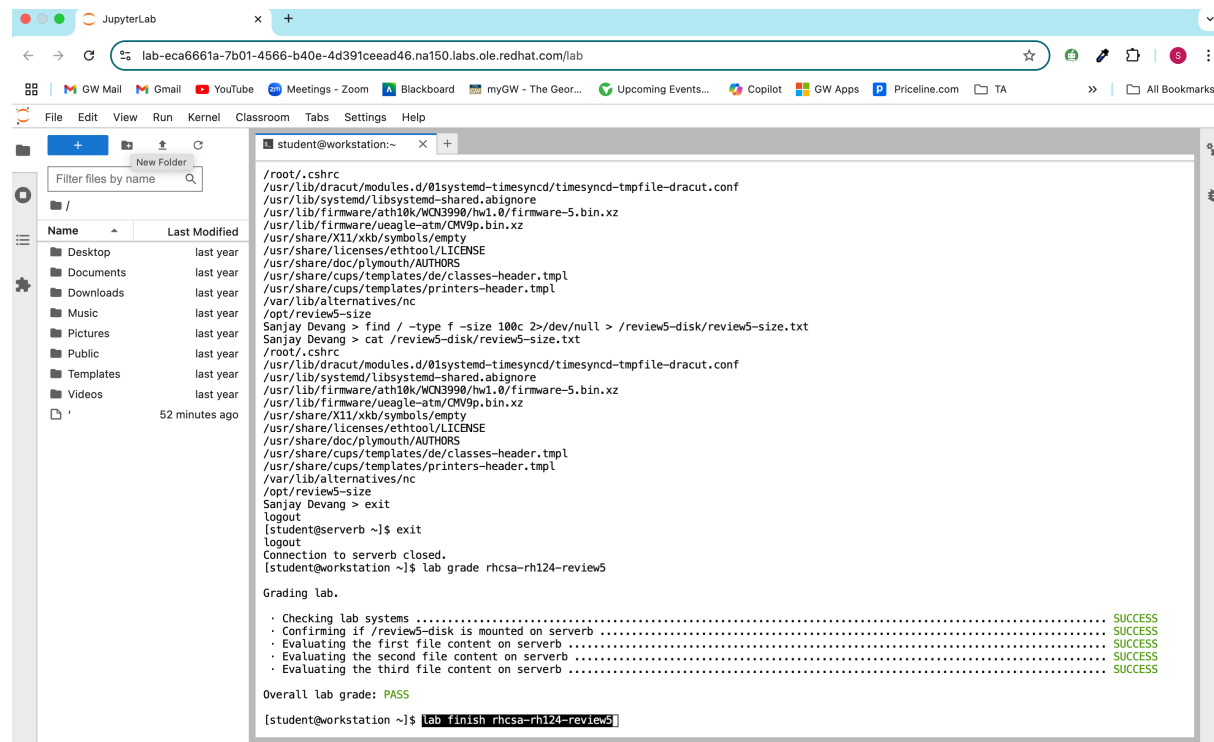
Figure 17 Figure 15 Configuring a static IP connection using nmcli and validating network settings in Red Hat Enterprise Linux.

Note: Identified the correct UUID for /dev/vdb1 using lsblk and mounted it at /mnt/freespace. Verified mount status, created disk usage and search result files, and successfully completed all tasks for the fs-review lab. The grading script confirmed correct mount, disk usage check.

Project 1 – RH124 Linux Administration I

SANJAY DEVANG

ASSIGNMENT -4



```
student@workstation:~  
/root/.cshrc  
/usr/lib/dracut/modules.d/01systemd-timesyncd/timesyncd-tmpfile-dracut.conf  
/usr/lib/systemd/libsystemd-shared.abignore  
/usr/lib/firmware/ath10k/WCN3990/hw1.0/firmware-5.bin.xz  
/usr/lib/firmware/ueagle-atm/OMV9p.bin.xz  
/usr/share/X11/xkb/symbols/empty  
/usr/share/licenses/ethtool/LICENSE  
/usr/share/doc/plymouth/AUTHORS  
/usr/share/cups/templates/de/classes-header.tmpl  
/usr/share/cups/templates/printers-header.tmpl  
/var/lib/alternatives/nc  
/opt/review5-size  
Sanjay Devang > find / -type f -size 100c 2>/dev/null > /review5-disk/review5-size.txt  
Sanjay Devang > cat /review5-disk/review5-size.txt  
/root/.cshrc  
/usr/lib/dracut/modules.d/01systemd-timesyncd/timesyncd-tmpfile-dracut.conf  
/usr/lib/systemd/libsystemd-shared.abignore  
/usr/lib/firmware/ath10k/WCN3990/hw1.0/firmware-5.bin.xz  
/usr/lib/firmware/ueagle-atm/OMV9p.bin.xz  
/usr/share/X11/xkb/symbols/empty  
/usr/share/licenses/ethtool/LICENSE  
/usr/share/doc/plymouth/AUTHORS  
/usr/share/cups/templates/de/classes-header.tmpl  
/usr/share/cups/templates/printers-header.tmpl  
/var/lib/alternatives/nc  
/opt/review5-size  
Sanjay Devang > exit  
logout  
[student@serverb ~]$ exit  
logout  
Connection to serverb closed.  
[student@workstation ~]$ lab grade rhcsa-rh124-review5  
  
Grading lab.  
  
• Checking lab systems ..... SUCCESS  
• Confirming if /review5-disk is mounted on serverb ..... SUCCESS  
• Evaluating the first file content on serverb ..... SUCCESS  
• Evaluating the second file content on serverb ..... SUCCESS  
• Evaluating the third file content on serverb ..... SUCCESS  
  
Overall lab grade: PASS  
[student@workstation ~]$ lab finish rhcsa-rh124-review5
```

Figure 18 Completing Review5 Lab: Filesystem Mounting and File Search in Red Hat Enterprise Linux

Note: Mounted the unmounted XFS filesystem `/dev/vdb1` to `/review5-disk` and created three result files using `find` for path location, ownership & permission filtering, and exact file size matching. The grading script confirmed correctness of all three tasks and mount status.